AMD DEVELOPER HACKATHON: ACT III · INTELLIGENT INDUSTRY

worksight.

Few-shot visual inspection for quality engineers

Onboard a part from 10–20 good photos. Get a heatmap, the closest good reference and an explanation from **Gemma on AMD**. The engineer makes the call.

Live: **web-production-d9743.up.railway.app**
Code: **github.com/DesertEaglesTech/worksight** · MIT



Real test image: VisA pcb1, bent pins (Amazon, CC BY 4.0)

THE PROBLEM

A new part on the line needs inspection today, not after a defect dataset exists

WHO

The quality engineer

Owns the pass / fail decision for a part and camera view. Has plenty of photos of good parts and very few of defects.

TODAY

Supervised models wait for defects

A typical defect classifier needs many labelled defect images and a training run for every new part. Defects are rare, so the data arrives last.

AND

A score alone is not evidence

The engineer has to see **where** the part differs, **compared with what**, and record their own decision.

THE PRODUCT

Onboard, inspect, explain, decide

1 · ONBOARD

~9 s

Profile from 10–20 photos of good parts. No defect labels, no training run. Measured: 20 references on Railway CPU.

→

2 · INSPECT

0.1–0.3 s

Anomaly score, heatmap of unusual regions and the most similar good reference. Normal / review / anomaly.

→

3 · EXPLAIN

Gemma

Gemma-4-31B-IT on AMD Token Factory describes what looks wrong and what to check, using the part's documentation.

→

4 · DECIDE

Human

The engineer records their decision and note. It is stored separately; the model result is never overwritten.

Live check on the hosted app: VisA pcb1 “bent pins” → **Fail**, score 0.279 against threshold 0.094, heatmap on the pins, Gemma description returned.

APPLICATION OF TECHNOLOGY

Gemma on AMD explains every inspection

Browser · Next.js

Operator workspace. A server-side proxy keeps the API key out of the browser.

FastAPI · PyTorch detector

Frozen ResNet-18 patch memory, nearest-patch retrieval, heatmap. ROCm-aware device selection.

AMD Token Factory · Gemma-4-31B-IT

Gets the photo, heatmap overlay, closest good reference, detector decision and part documentation. Fallback: DeepSeek-V4.1-Flash, also on Token Factory.

- **Grounded:** Gemma reads the same evidence the engineer sees, plus the part's own PDF or text documentation.
- **Bounded:** the detector's decision and score are passed in and never changed. The report is labelled assistive.
- **Traceable:** each report records which model answered and its response time.
- **Resilient:** two retries on server errors, then the fallback model.

Next Detector inference on an AMD GPU: **scripts/amd_run.sh** runs the smoke test and benchmark with --require-amd. Not run yet; hosted detector is CPU.

MEASURED RESULTS

Real boards, held-out test, no test-set tuning

VISA	AUROC	RECALL	FALSE ALARMS	P50 LATENCY
pcb1	0.905	44%	5%	250 ms
pcb2	0.884	59%	6%	304 ms
pcb3	0.875	46%	4%	250 ms
pcb4	0.992	87%	2%	194 ms

20 references from the official train normals; full official test split (100 normal + 100 defect images per board). Design chosen on 8 other VisA categories. CPU run; not an AMD run.

LIVE SPOT CHECK

10-image pcb1 batch on the hosted app: **all 5 defects flagged**, one good board sent to **review**.

HONEST LIMITS

Recall is ~45% on pcb1/pcb3 at this threshold. The score is a distance, not a probability: Worksight is a triage tool with a review band, not an autonomous rejecter.

Source: docs/BENCHMARK.md, docs/STATUS.md

HUMAN + AI WORKFLOW

The engineer stays in charge of every change

FEEDBACK

Engineer confirms or rejects the model result. Saved feedback is immutable.

→

CANDIDATE

Confirmed-normal parts build a new profile version. It is not active yet.

→

GATE

Compared with the current version on a frozen, labelled validation set. Any new false positive or false negative blocks it.

→

ACTIVATE

Explicit activation, audit record, rollback to an earlier approved version.

Old inspections keep the version they used, so every past decision can be traced to the exact references behind it.

BUSINESS VALUE

Inspection from day one of a new part

- **No defect dataset to start:** a part is onboarded from good photos the line already has, in seconds instead of a labelling and training cycle.
- **Engineer time on the right parts:** results are sorted into normal, review and anomaly; batch inspection ranks a folder by score through the API.
- **Evidence for every decision:** heatmap, closest reference, Gemma explanation and the engineer's own note, stored together.
- **Safe updates:** a profile only changes after a measured gate and explicit approval.

BUILT

Live web app, FastAPI backend, Gemma reports, versions with gate and rollback. 36 backend + 39 web tests.

NEXT

Detector on AMD Instinct via ROCm, batch screen in the UI, pilot on a real line's camera.